Lesson Plan : 22th May 2025

Students prepare for **Test 2 (Lab Test)**, covering both **Part A: Debugging** and **Part B: Programming**.

This lesson plan also includes sample exercises and a UML class diagram for a typical OOP-based problem using Association, Aggregation, and Composition.

Lesson Plan: Java Programming - Test 2 Preparation Class

Objective:

To prepare students for Test 2 (Lab Test) through targeted exercises and guidance covering Java fundamentals, ArrayLists, and Class Relationships (Association, Aggregation, Composition).

Test 2 Overview:

- **Mode:** Lab Test (Individual)
- **Parts:**
    - **Part A: Debugging**
        - Students will be given `.java` files with syntactic/semantic errors.
        - Goal: Fix the errors and get the program running correctly.
    - **Part B: Programming**
        - Students will be given a programming problem (with a UML class diagram).
        - Goal: Implement Java classes using the correct structure, relationships, and logic.
- **Focus Chapters:**

    - **Chapter 2:** Classes and Objects (Constructors, Data Types, Methods)
    - **Chapter 5:** ArrayList
    - **Chapter 6:** Class Relationships (Association, Aggregation, Composition)
- **Tools Provided:** `.java` skeleton files
- **Submission Method:** Via Pen Drive (not through e-learning)

Exercise Samples

✅ Part A: Debugging Exercise Sample

Buggy File: `Patient.java`

```java
public class Patient {
    private String name;
    private int age

    public void Patient(String name, double age){
        name = name;
        age = age;
    }

    public String displayDetails() {
        System.out.println("Name: " + name)
        System.out.println("Age: " + age);
    }
}
```

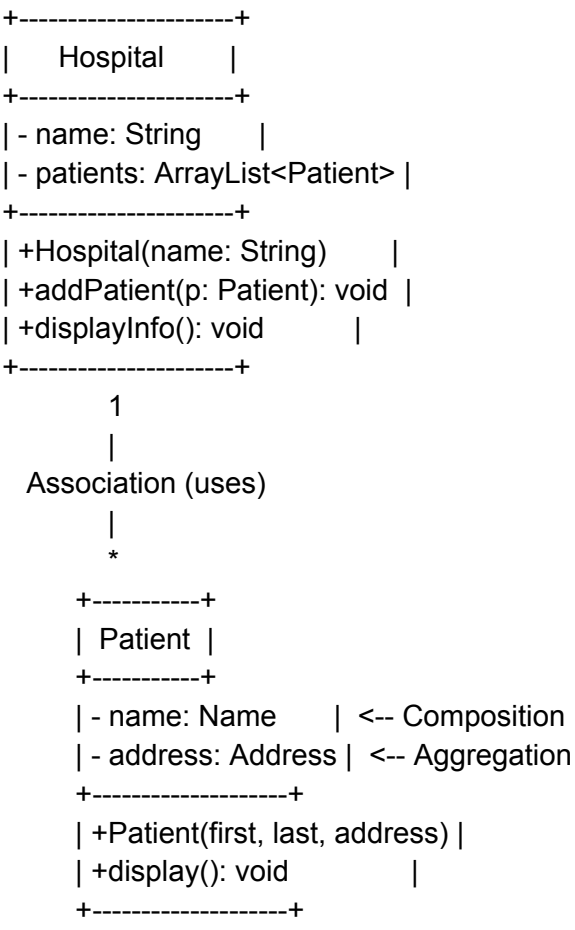
✅ **Student Task**: Identify and fix the bugs (e.g., missing semicolons, wrong variable assignment, etc.)

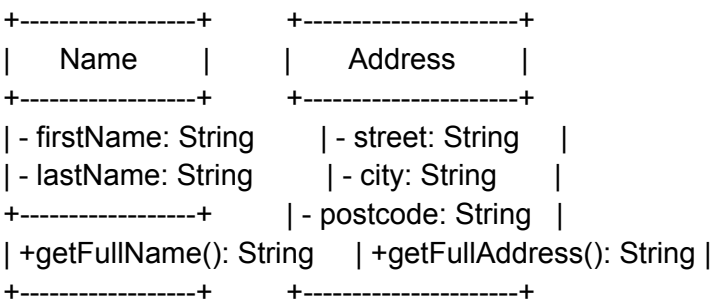
✅ Part B: Programming Exercise Sample

Problem Statement:

Implement a hospital management system using Association, **Aggregation** and **Composition** concepts. The system should include:

- **Hospital**
- **Patient**
- **Name** (firstName, lastName)
- **Address** (street, city, postcode)
- **Ward** (shared between Hospital and Patients)






```
Unset
```

✅ **Student Task**: Implement the above classes in Java using appropriate relationships, constructors, and methods.

Key Reminders for Students:

- All files will be given in `.java` format.
- Focus on:
    - Proper use of constructors
    - Access modifiers
    - Relationship mapping (has-a)
    - Using `ArrayList` for multiple objects
- Submission **only via Pen Drive** at the end of the session.
- Ensure code compiles and runs before submitting.